

Internship Report

On

Data Engineering Virtual Internship

(AWS Academy Curriculum, under AICTE – EduSkills National Internship Portal)

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE (Core)



By

Name of Student: DISHA KUMARI

Roll Number / Enrollment Number: 2410031276

Section: 3CSE24

Semester: 5th Semester

Session: 2024-2028

CANDIDATE'S DECLARATION

I, DISHA KUMARI, Roll No. 2410031276, do hereby declare that the internship report titled “Data Engineering Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: DISHA KUMARI

Roll No.: 2410031276

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful for the structured learning and practical exposure provided through the internship program. I also express my gratitude to my parents for their blessings.

Signature: _____

Student Name: DISHA KUMARI

Roll No.: 2410031276

INTERNSHIP COMPLETION CERTIFICATE



AWS Academy Certificate of Completion — AWS Academy Graduate - Data Engineering - Training Badge (40 hours), issued 06/02/2026.



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EduSkills®
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Certificate of Virtual Internship

This is to certify that

Disha Kumari

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Engineering Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Curriculum Provided by:



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 45df8c17ed649c512df6

Student ID: STU6831912c2a6bb1748078892



TABLE OF CONTENTS

Chapter	Chapter Name	Page
1	Candidate's Declaration	i
2	Acknowledgement	ii
3	Internship Completion Certificate	—
4	Project Description	1
4.1	Introduction	1
4.2	Organization Profile	2
4.3	Problem Statement	2
4.4	Project Objectives	3
4.5	Scope of the Project	3
4.6	Technologies and Tools Used	4
4.7	System Architecture	5
4.8	Methodology	6
4.9	Expected Outcomes	8
4.10	Certificates and Communication Proof	9
5	Bibliography / References	10

4. PROJECT DESCRIPTION

4.1 Introduction

The internship undertaken by DISHA KUMARI was an 8-week AICTE–EduSkills Virtual Internship Program in the domain of AWS Data Engineering. The offer letter identifies the internship duration as April 2026 – June 2026 and describes the program as a structured learning experience combining weekly learning, assessments, project documentation/assignments and a final assessment.

The internship covered data-driven organizations and pipelines, data quality and pipeline design, security and ingestion, big-data processing, machine-learning preparation, analytics and visualization, automation, and certification preparation.

4.2 Organization Profile

The internship offer identifies EduSkills as the organization/program provider and states that the program is conducted under the AICTE–EduSkills Virtual Internship Program. The offer letter is signed by Bibek Ranjan, Director, EduSkills.

The supplied AWS Academy credential identifies DISHA KUMARI as an AWS Academy Graduate in Data Engineering – Training Badge, with 40 hours of course completion and an issue date of 06/02/2026.

4.3 Problem Statement

Modern organizations generate data from multiple sources and require reliable ways to ingest, store, transform, secure, analyze and visualize that data. A data-engineering workflow must address data volume, variety, velocity and quality while supporting scalable processing and dependable delivery.

The internship therefore focused on understanding how cloud-based data pipelines can be designed and operated using AWS services, including storage, querying, ETL, streaming, big-data processing, analytics and automation.

4.4 Project Objectives

- Understand the role of a data engineer in data-driven organizations.
- Learn principles of data elements, data quality and data-pipeline design.
- Understand cloud security, scaling, ETL/ELT and data-ingestion approaches.
- Gain familiarity with AWS services used for storage, querying, ETL and analytics.
- Understand batch and stream ingestion and modern big-data processing.
- Explore preparation of data for machine learning and the AWS ML lifecycle.

- Study analytics and visualization using AWS services.
- Understand automation, infrastructure deployment and CI/CD concepts.
- Bridge the learning experience toward AWS certification preparation.

4.5 Scope of the Project

The scope is limited to the learning modules and practical exposure explicitly described in the internship offer. It covers AWS data engineering, pipeline design, security, ingestion, ETL/ELT, big-data technologies, machine-learning preparation, analytics, visualization and automation.

The report does not claim an independent production system, commercial deployment, or specific performance benchmark because such details are not provided in the supplied documents.

4.6 Technologies and Tools Used

Amazon S3; Amazon Athena; AWS Glue; Amazon Kinesis; Kinesis Data Firehose; Amazon Redshift; Hadoop; Apache Spark; Amazon EMR; Amazon SageMaker; SageMaker Canvas; AWS IoT Analytics; Amazon QuickSight; AWS Step Functions; ETL/ELT, data quality, security, scaling and CI/CD concepts.

4.7 System Architecture

Conceptual training architecture: Data Sources → Ingestion (Kinesis / batch and stream ingestion) → Storage (Amazon S3 / Redshift) → Processing (AWS Glue / Athena / EMR-Spark) → Analytics & ML (QuickSight / IoT Analytics / SageMaker). Security, data quality, scalability and automation apply across the pipeline.

4.8 Methodology

The internship followed the week-wise structured learning plan stated in the offer letter. The methodology combined guided learning, hands-on labs, weekly assessments, project documentation/assignments and a final assessment.

Week 1: Data Engineering for Data-Driven Organizations — pipelines, role of the data engineer and S3 practice.

Week 2: Data Elements and Pipeline Design Principles — five Vs, data quality, architectures and Athena.

Week 3: Securing, Scaling, and Ingesting Data Pipelines — security, scaling, ETL vs ELT and data wrangling.

Week 4: Efficient Data Management & Big Data — batch/stream ingestion, Glue, Kinesis and Redshift.

Week 5: Big Data & Preparing Data for Machine Learning — Hadoop, Spark, EMR, SageMaker and Canvas.

Week 6: Analyzing and Visualizing Data — IoT Analytics, QuickSight and Kinesis Data Firehose.

Week 7: Automating the Pipeline — infrastructure deployment, CI/CD, Step Functions and Athena ETL labs.

Week 8: Bridging to Certification — AWS certification tracks and preparation resources.

4.9 Expected Outcomes

- Structured understanding of cloud-based data engineering workflows.
- Familiarity with AWS data services.
- Improved understanding of data quality, security and scalability.
- Awareness of Hadoop, Spark and EMR.
- Understanding of preparing data for machine learning and the AWS ML lifecycle.
- Exposure to visualization and analytics.
- Understanding of automation, orchestration and CI/CD.
- Greater readiness for further AWS certification-oriented learning.

4.10 Certificates of Completion and Communication Proof

The supplied AWS Academy credential is attached as evidence of 40 hours of Data Engineering training.

The AICTE–EduSkills Internship Offer Letter (Ref. C16Y2026M051421160) is attached as official internship offer and communication proof. It specifies the internship duration as April 2026 – June 2026 and states that there is no stipend associated with the internship.

The report format requires completion certificate and communication proof to be attached. If a separate final internship completion certificate has been issued by EduSkills, it should also be included in the final university submission package.

5. BIBLIOGRAPHY / REFERENCES

1. AICTE–EduSkills, Internship Offer Letter, Ref. No. C16Y2026M051421160, 2026.
2. AWS Academy, “AWS Academy Graduate - Data Engineering - Training Badge”, DISHA KUMARI, 40 hours, issued 06/02/2026.
3. IILM University, Internship Report Format and Guidelines, School of Computer Science and Engineering, 2026–27.